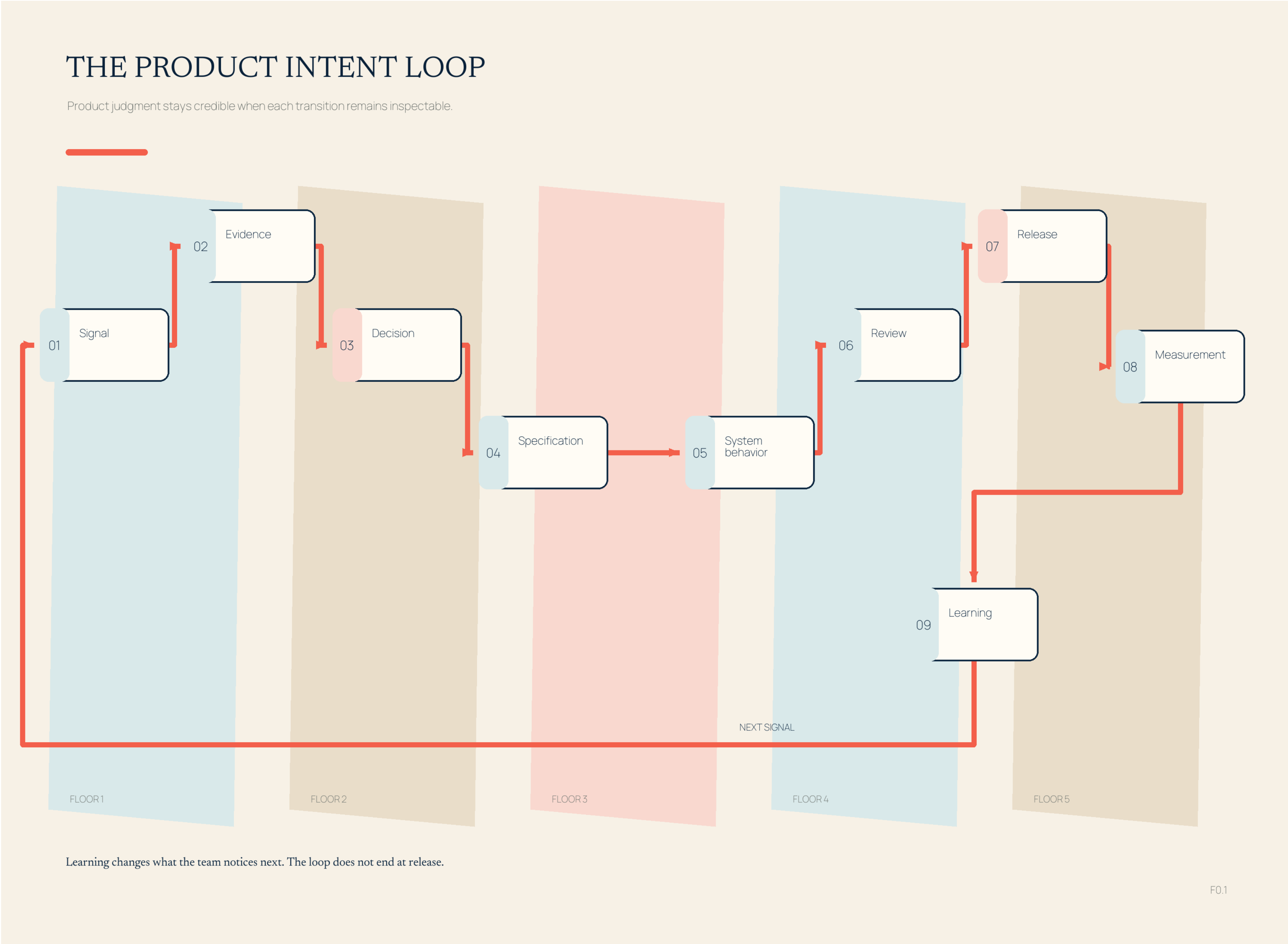
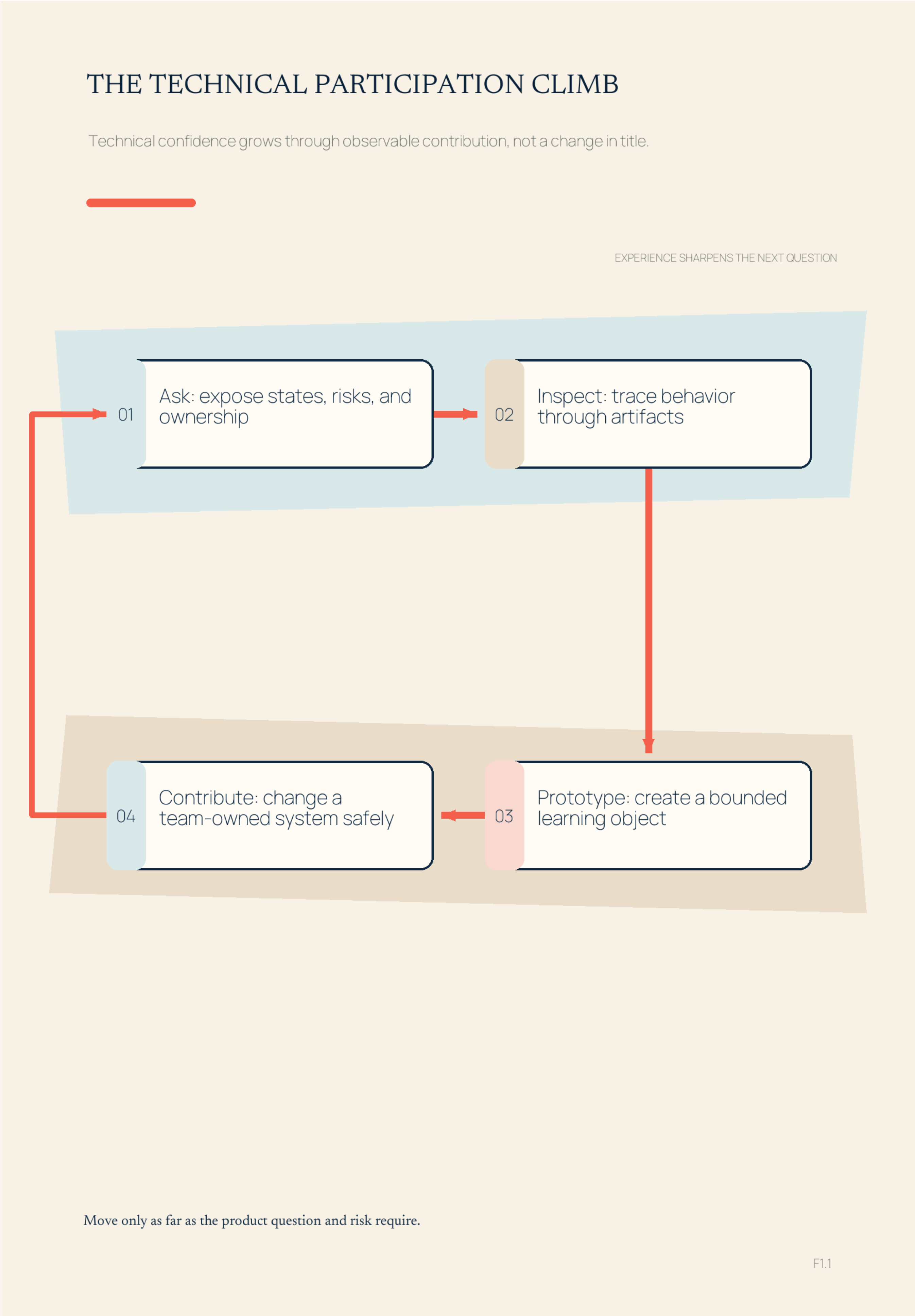
Figure 0.1. Signal-to-learning operating chain



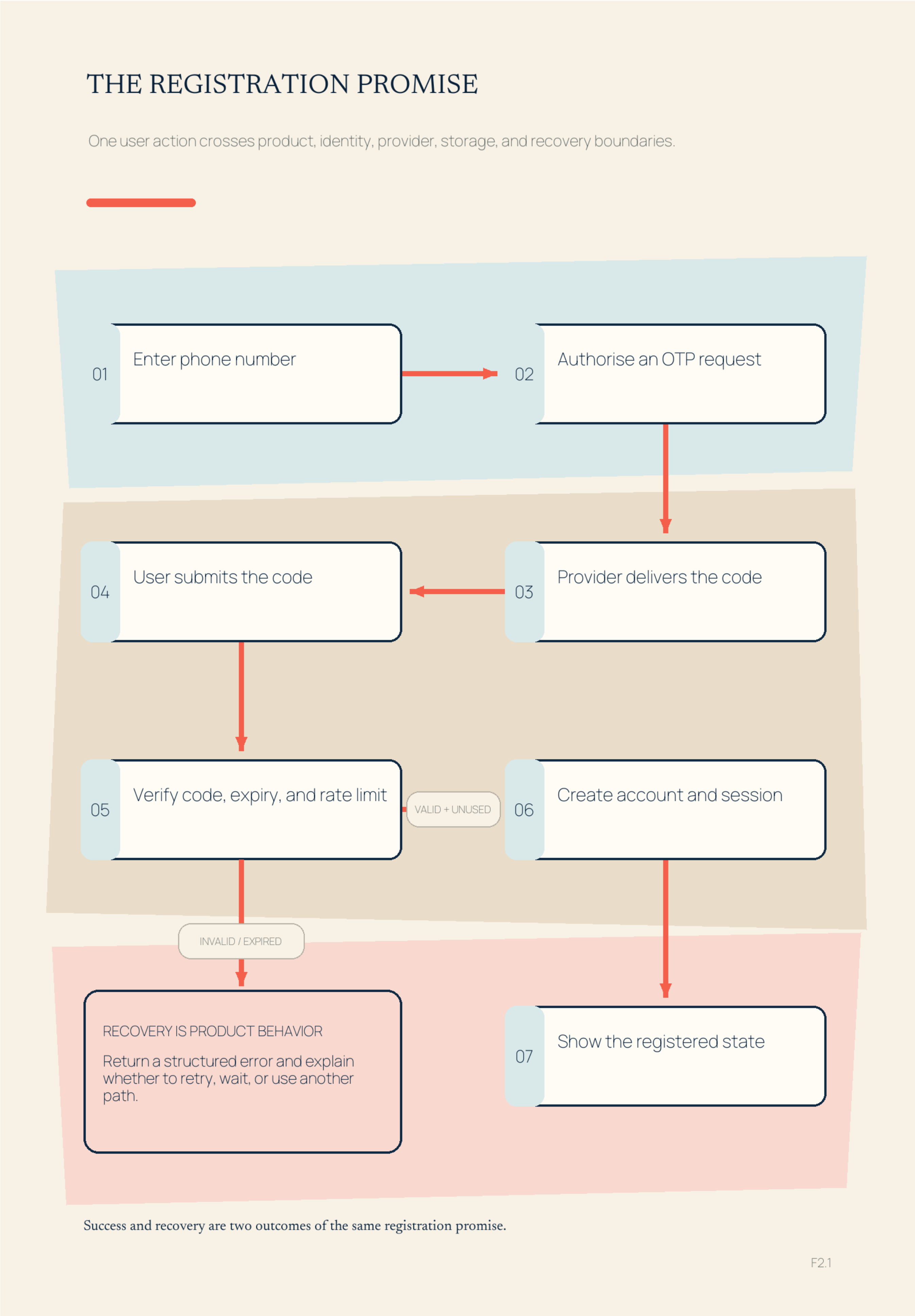
Product intent survives only when each transition remains inspectable; learning begins the next loop rather than ending the work.

Figure 1.1. Participation ladder: ask, inspect, prototype, contribute



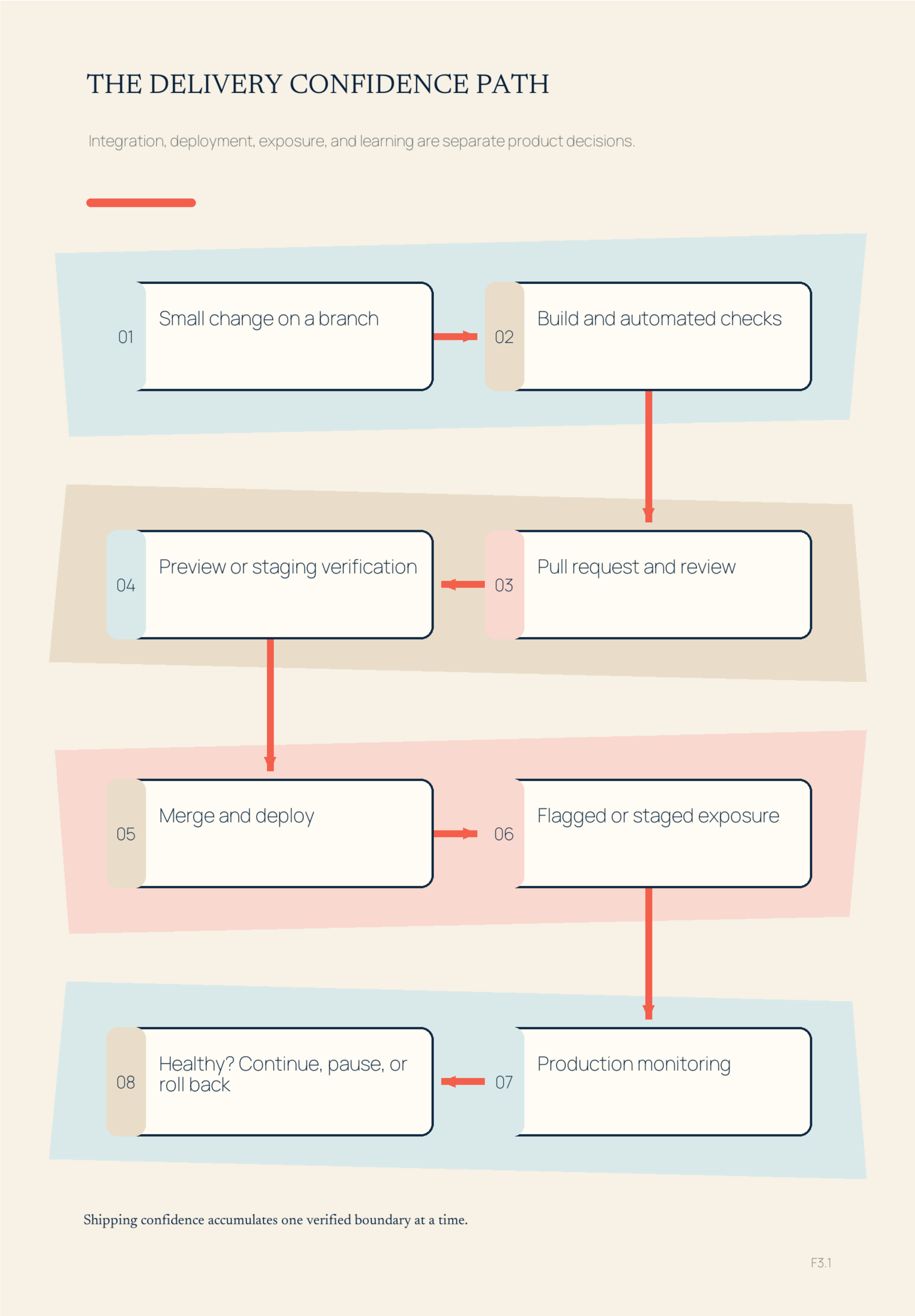
Technical participation progresses through observable work, not status; the appropriate stopping point depends on the product and risk.

Figure 2.1. Registration request/response sequence



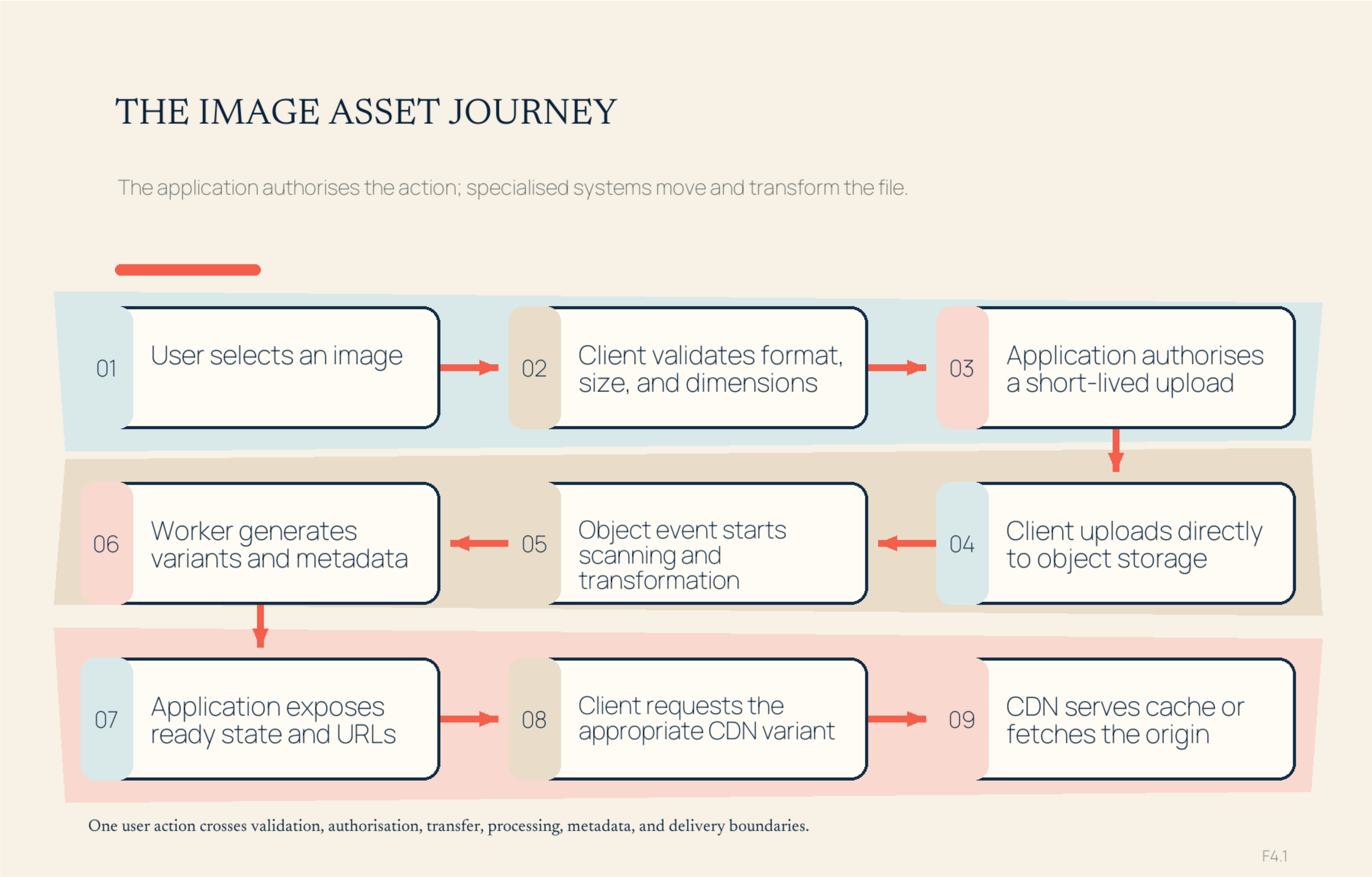
The registration promise crosses multiple actors; success and recovery states both belong in the product conversation.

Figure 3.1. Commit-to-production delivery path



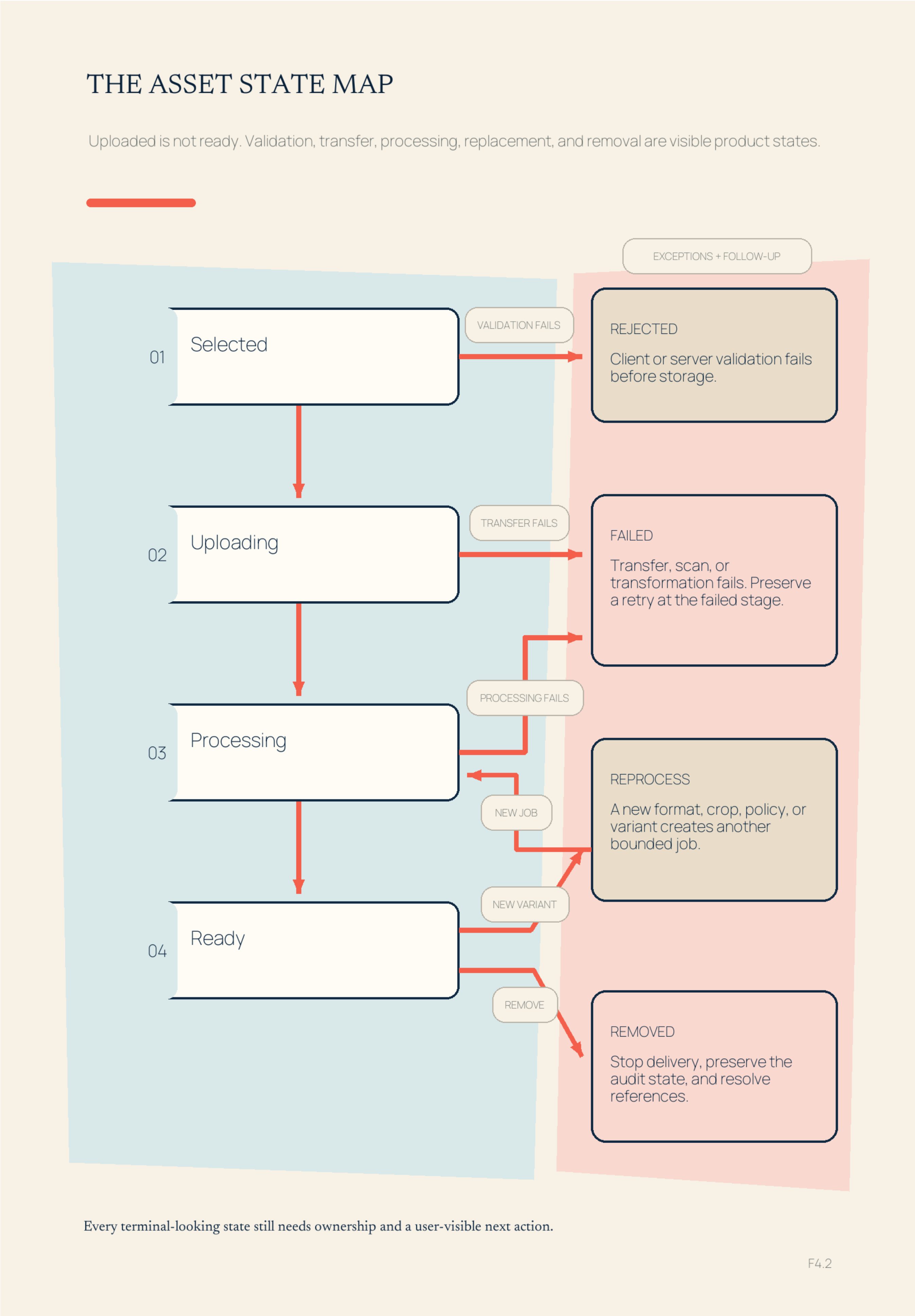
A delivery path separates code integration, deployment, exposure, and the decision to continue.

Figure 4.1. Original asset to responsive delivery



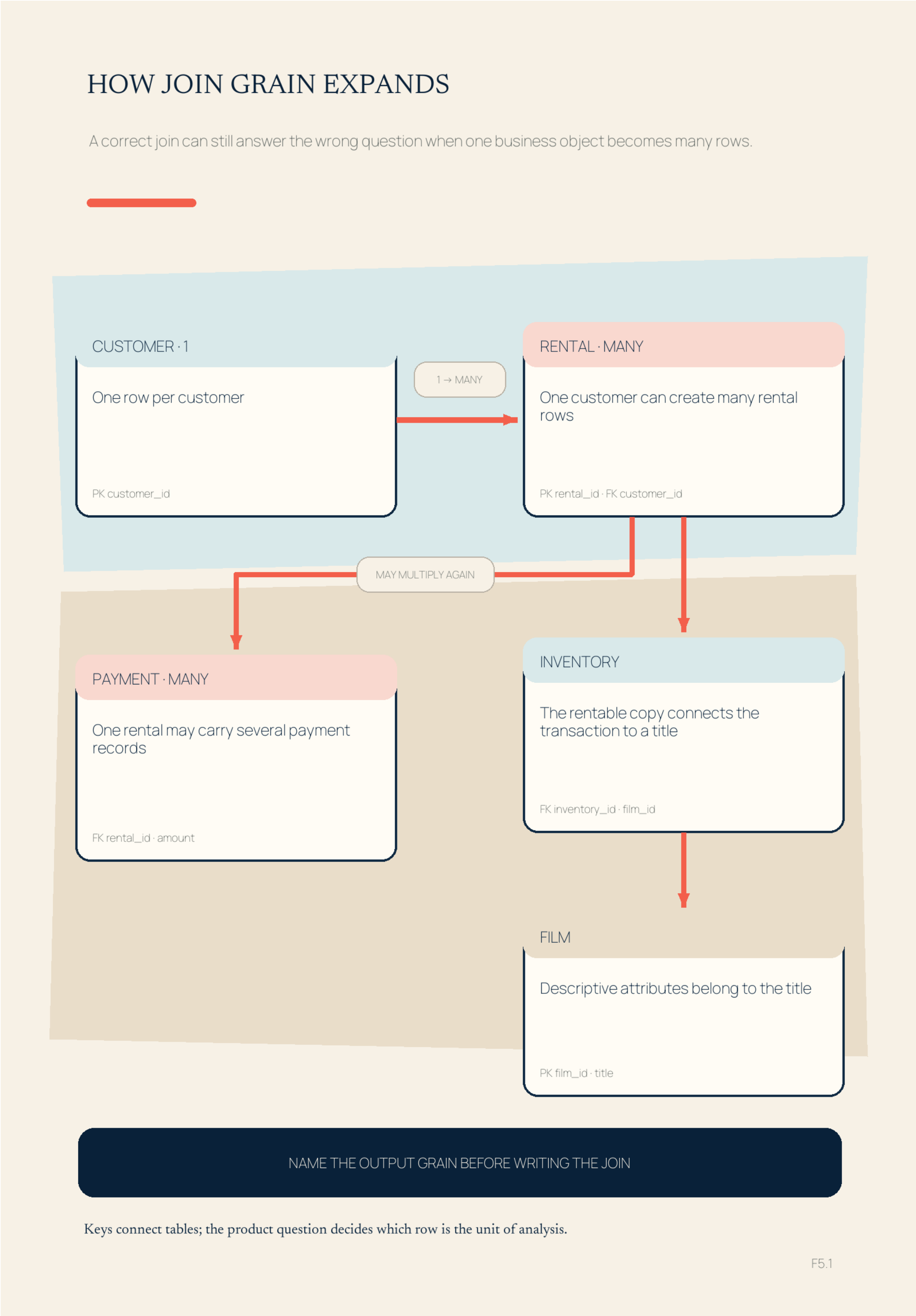
The image-asset journey connects selection and validation to authorised transfer, processing, product state, variants, and delivery. Later sections examine each boundary.

Figure 4.2. Asset-state lifecycle



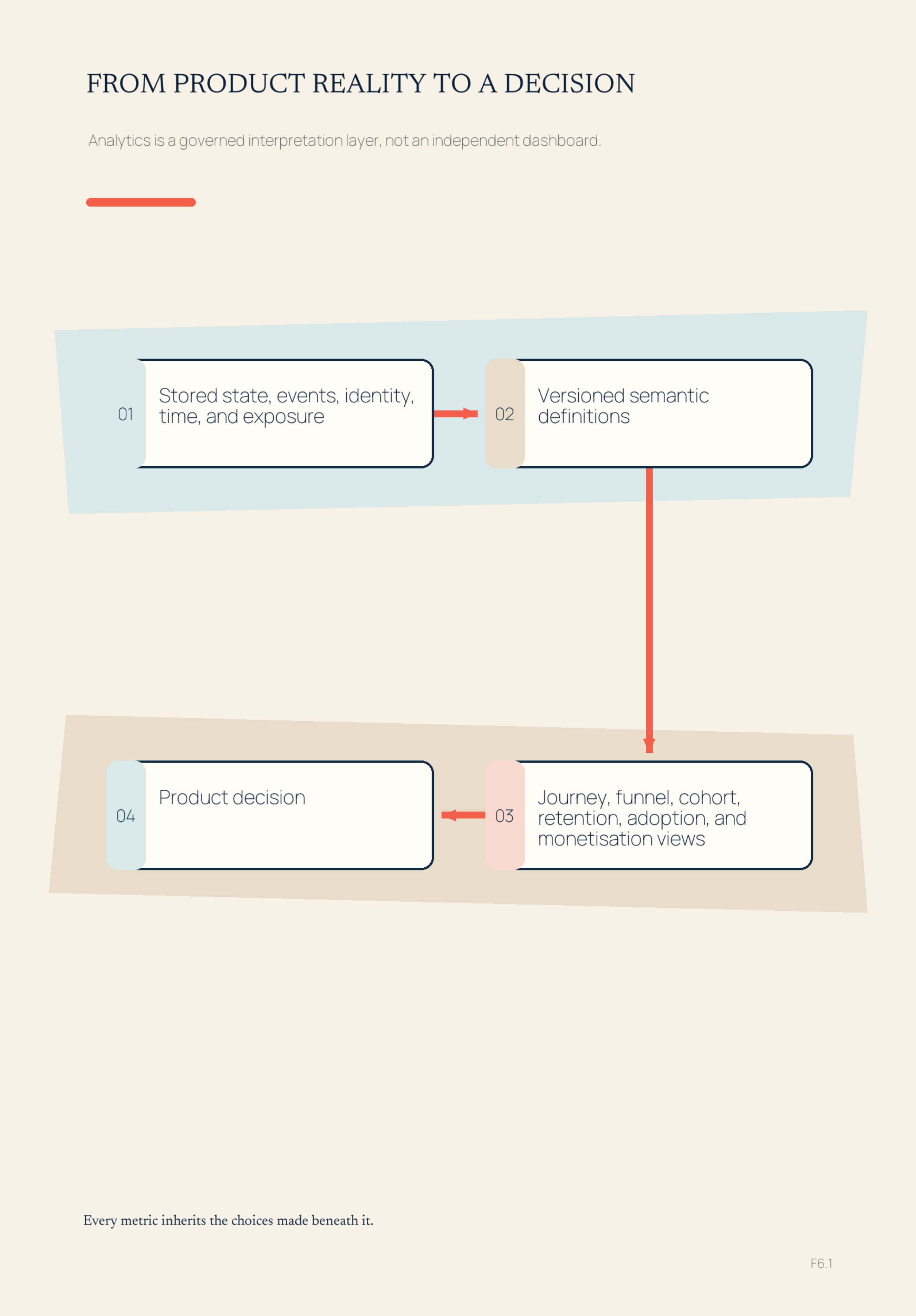
Asset state is user-visible product behavior; “uploaded” is not the only meaningful outcome.

Figure 5.1. Relational grain through joins



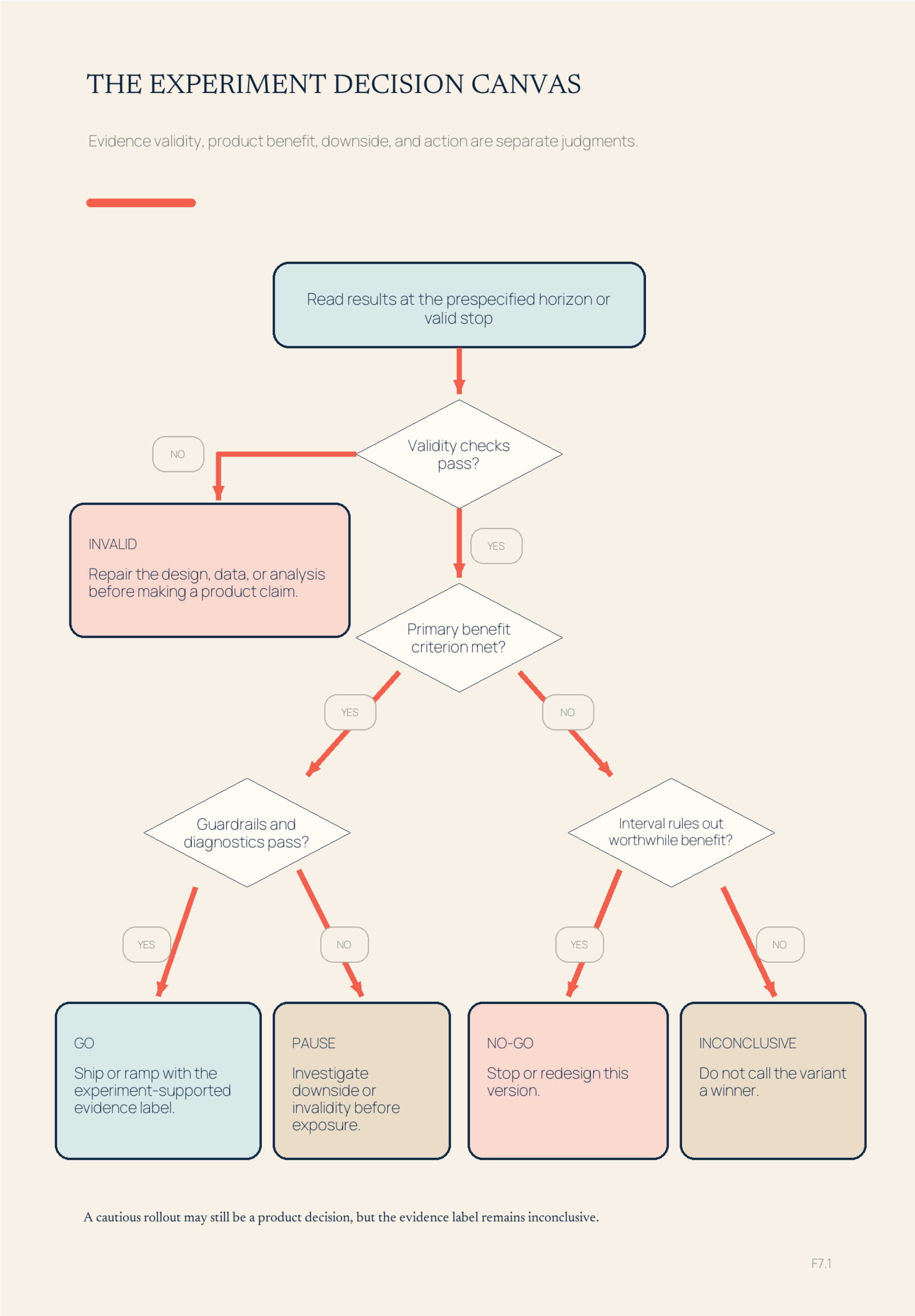
The query grain changes as tables join: one customer can have many rentals, and one rental may have multiple payment records. Confirm the intended unit before aggregating.

Figure 6.1. Event-to-metric semantic chain



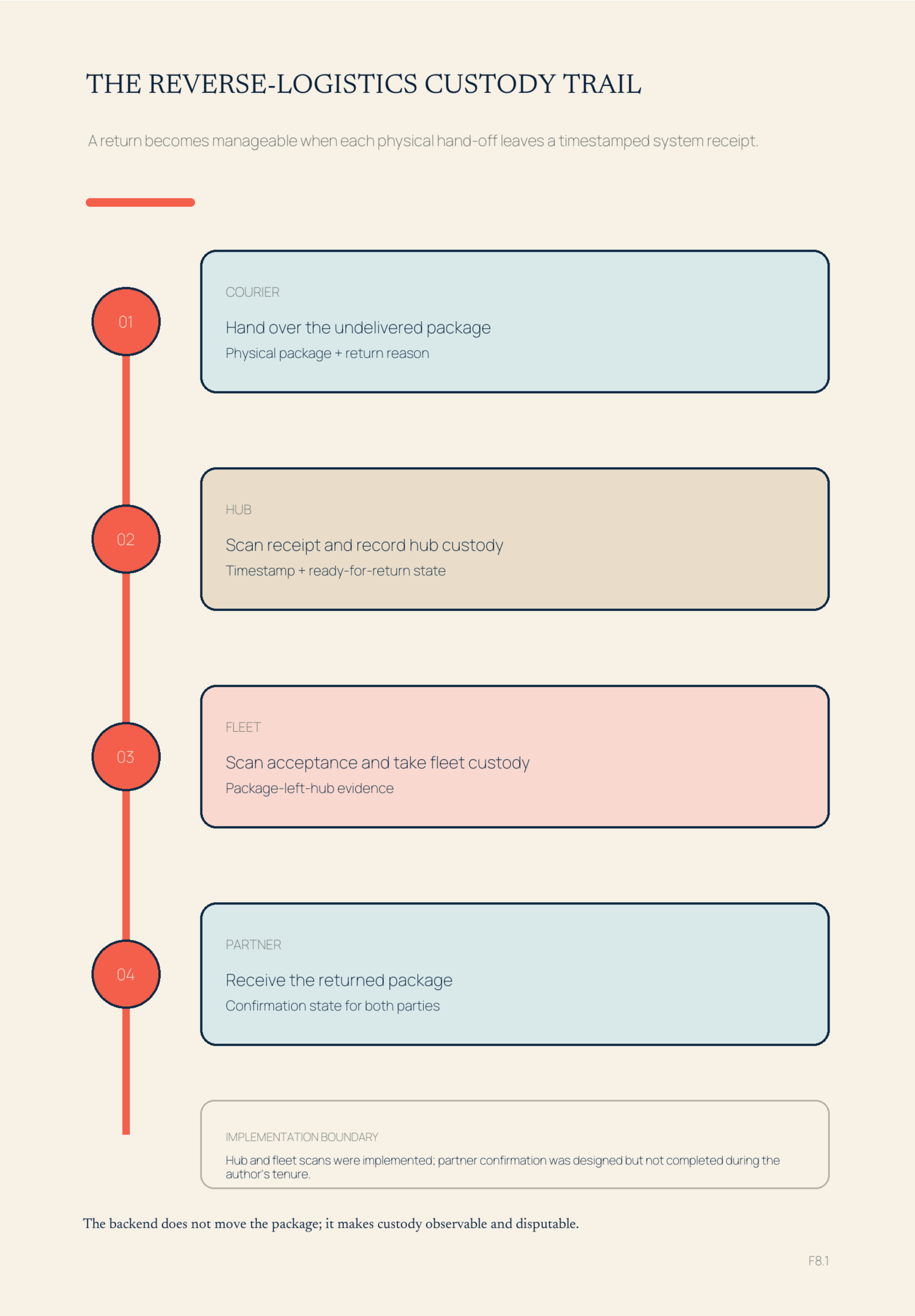
Product analytics combines stored state, event streams, identity, time, and semantic definitions; no dashboard exists independently of those choices.

Figure 7.1. Experiment go/no-go decision flow



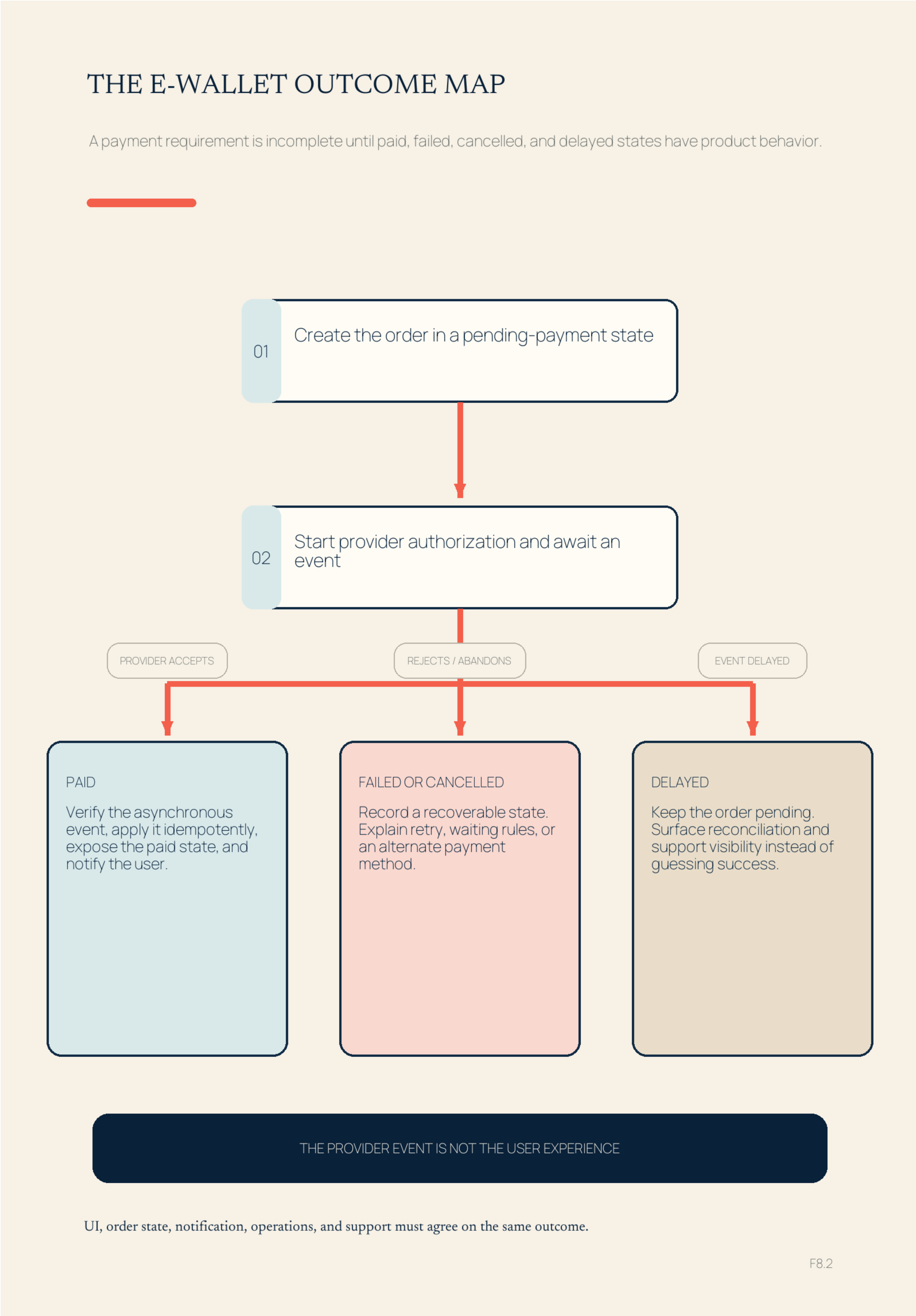
The go/no-go flow separates evidence validity, product benefit, downside, uncertainty, and action.

Figure 8.1. Shipper reverse-logistics custody sequence



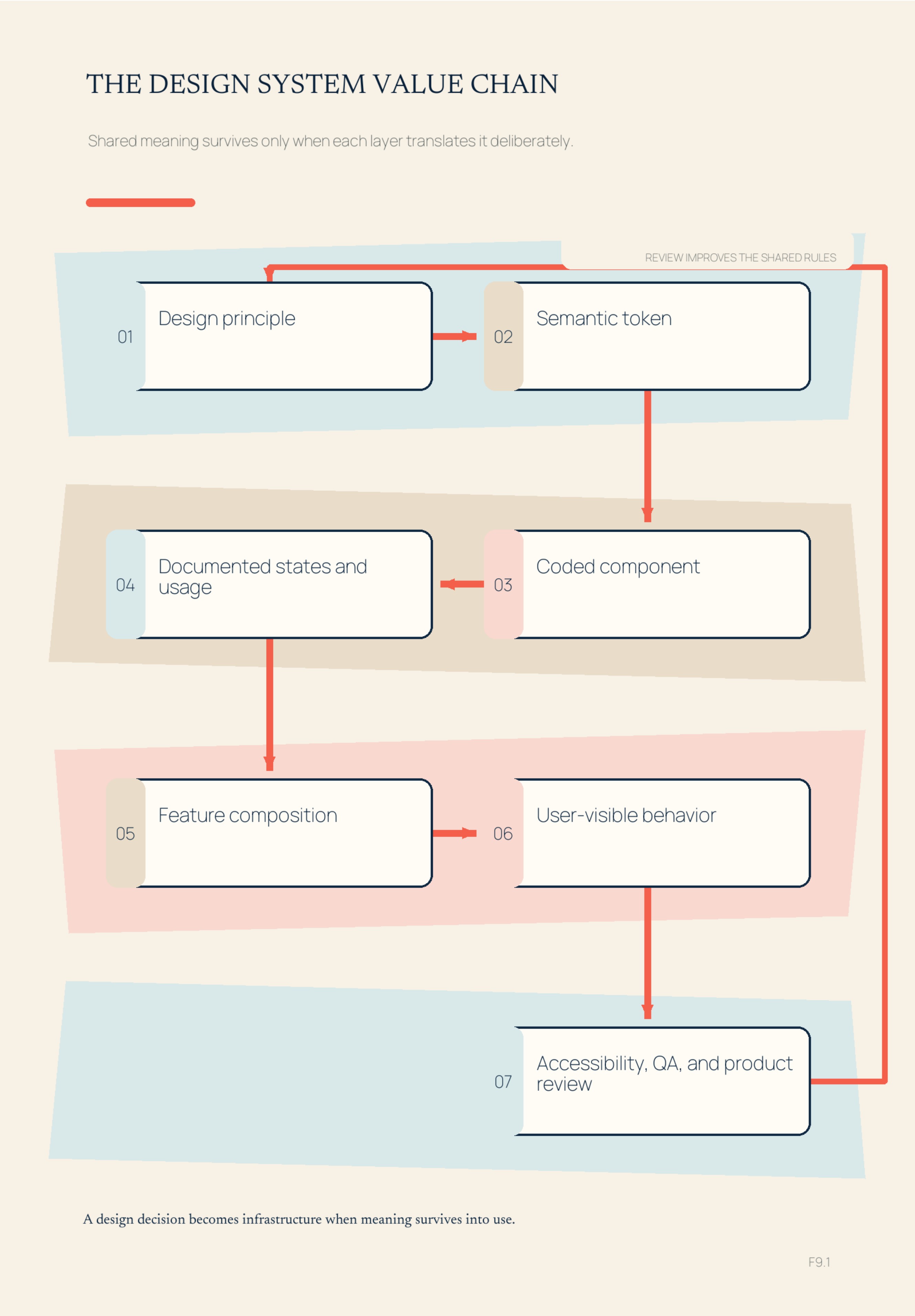
Publication-safe reconstruction of the reverse-logistics flow. Hub and driver scanning were implemented; partner-side receipt confirmation was designed but not completed during the author's tenure. Exact internal status names still require artifact or operational verification.

Figure 8.2. E-wallet flow with happy, error, and recovery states



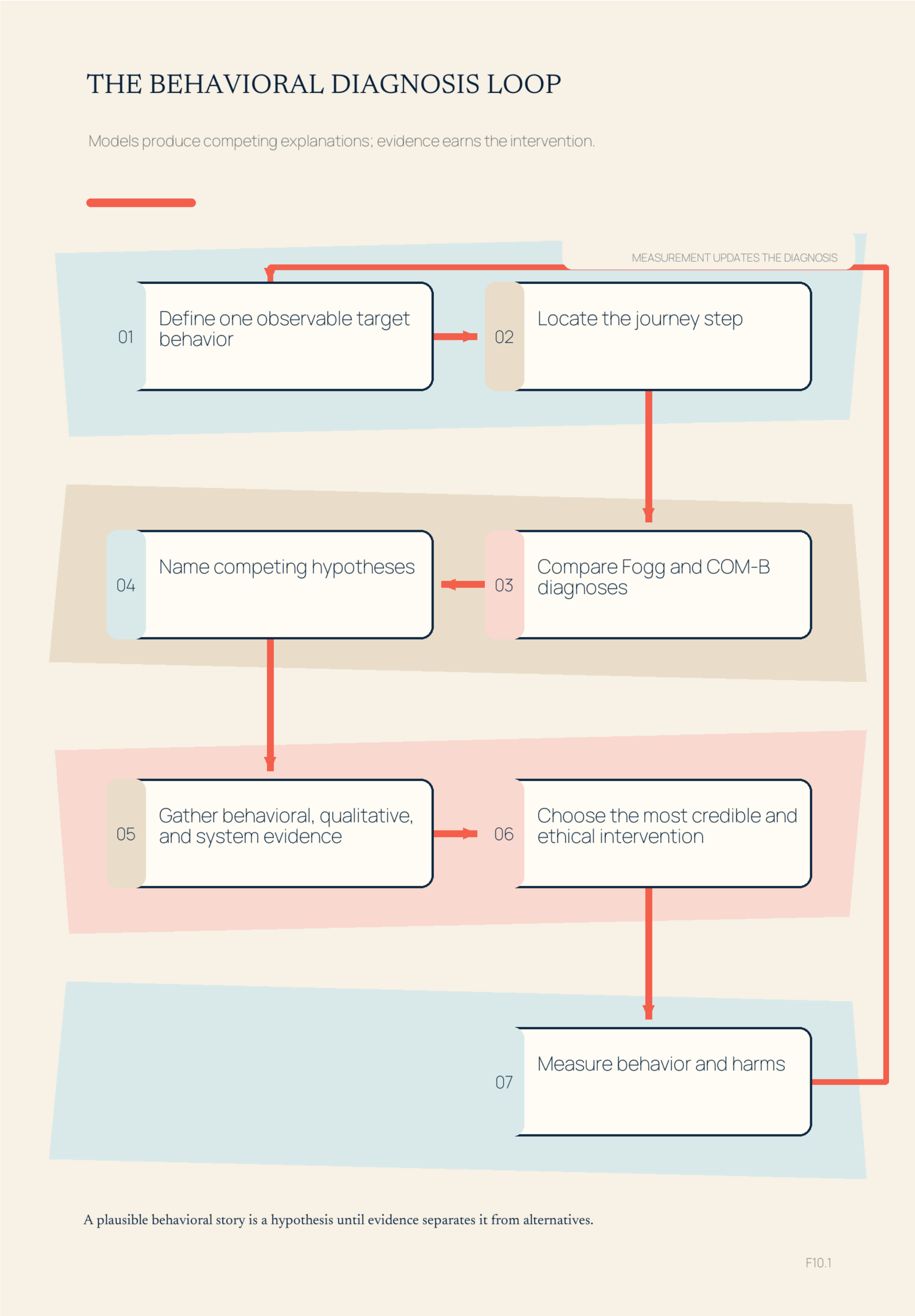
The e-wallet flow makes pending, failed, duplicate-safe, and support-visible behavior part of the requirement rather than late implementation detail.

Figure 9.1. Token-component-pattern-product hierarchy



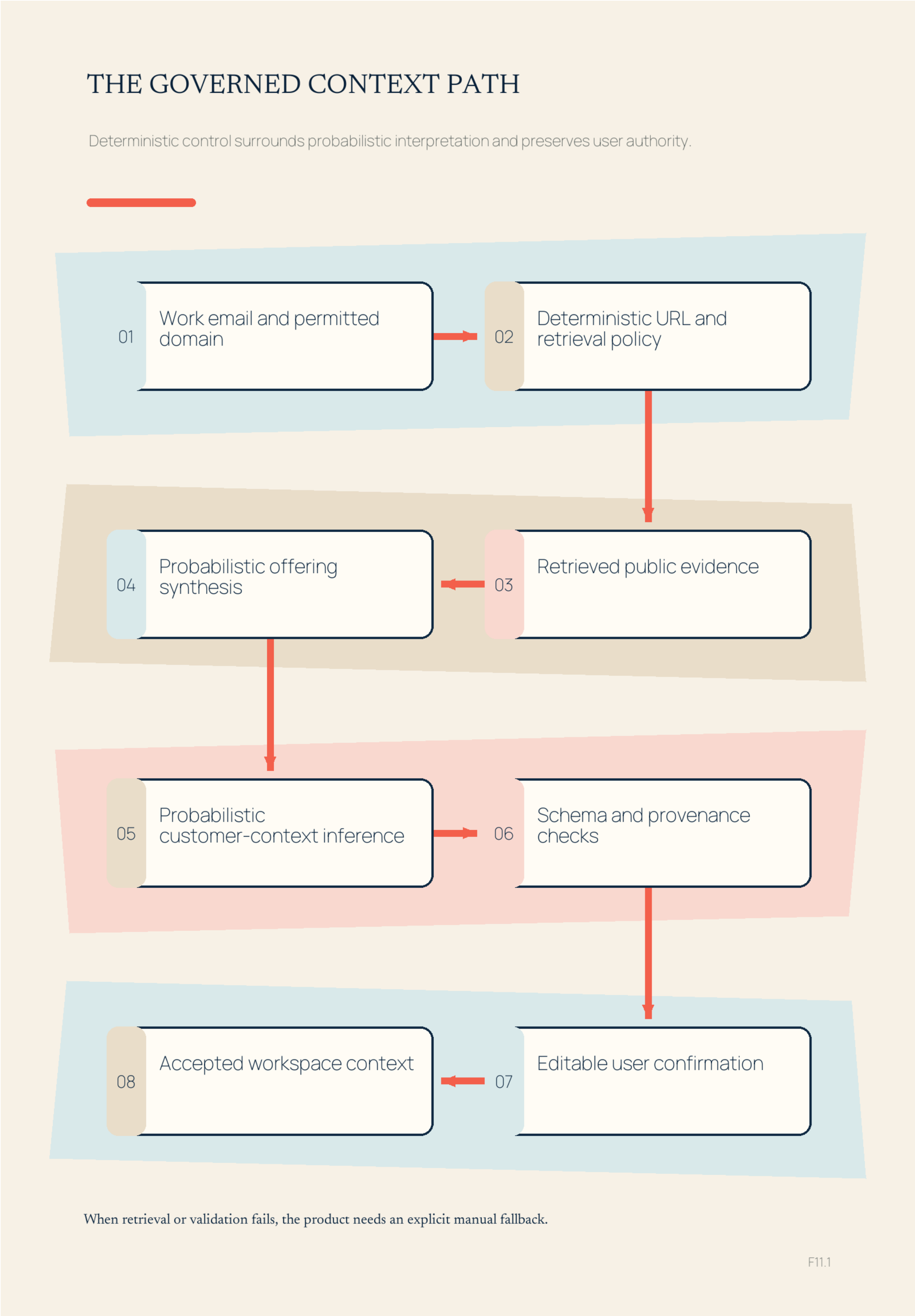
A design decision becomes infrastructure only when meaning survives from principle and token through code, states, composition, and review.

Figure 10.1. Behavior diagnosis: Fogg and COM-B



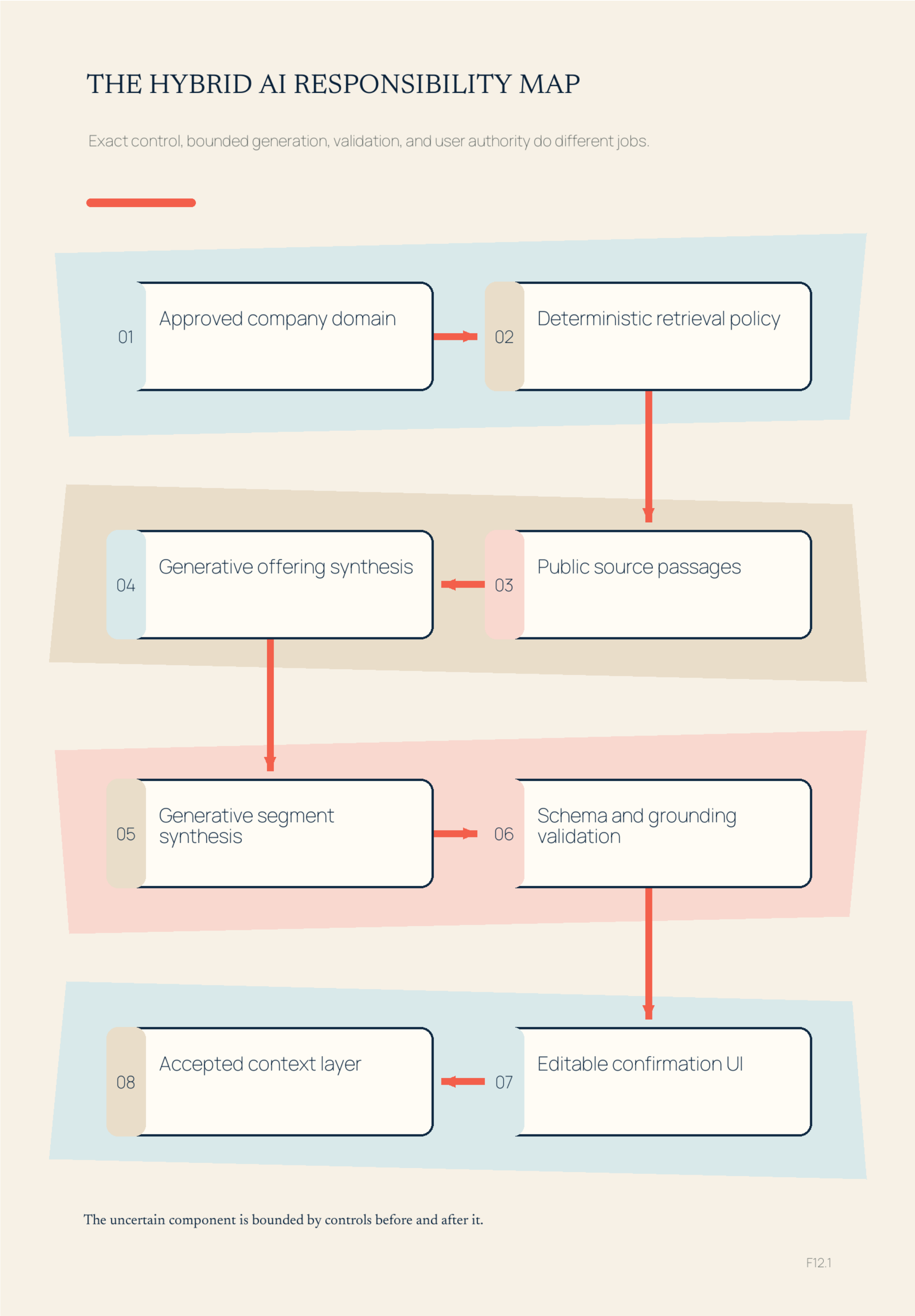
Behavioral models generate competing diagnoses; evidence selects among them, and measurement keeps the intervention accountable.

Figure 11.1. Heatseeker onboarding deterministic–probabilistic boundary



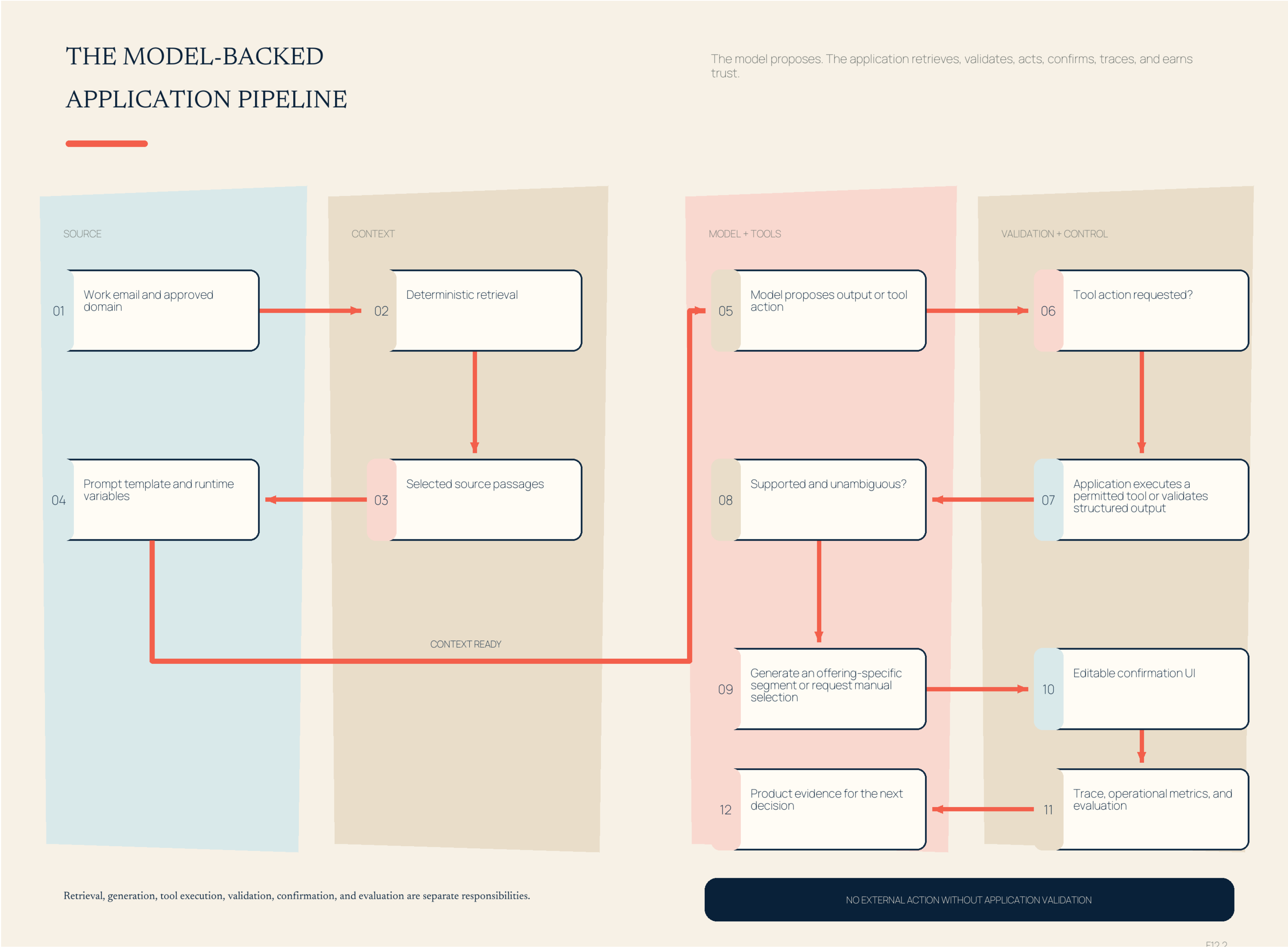
The capability becomes governable when deterministic controls, probabilistic interpretation, validation, fallback, and user authority are explicit.

Figure 12.1. Heatseeker hybrid solution shape



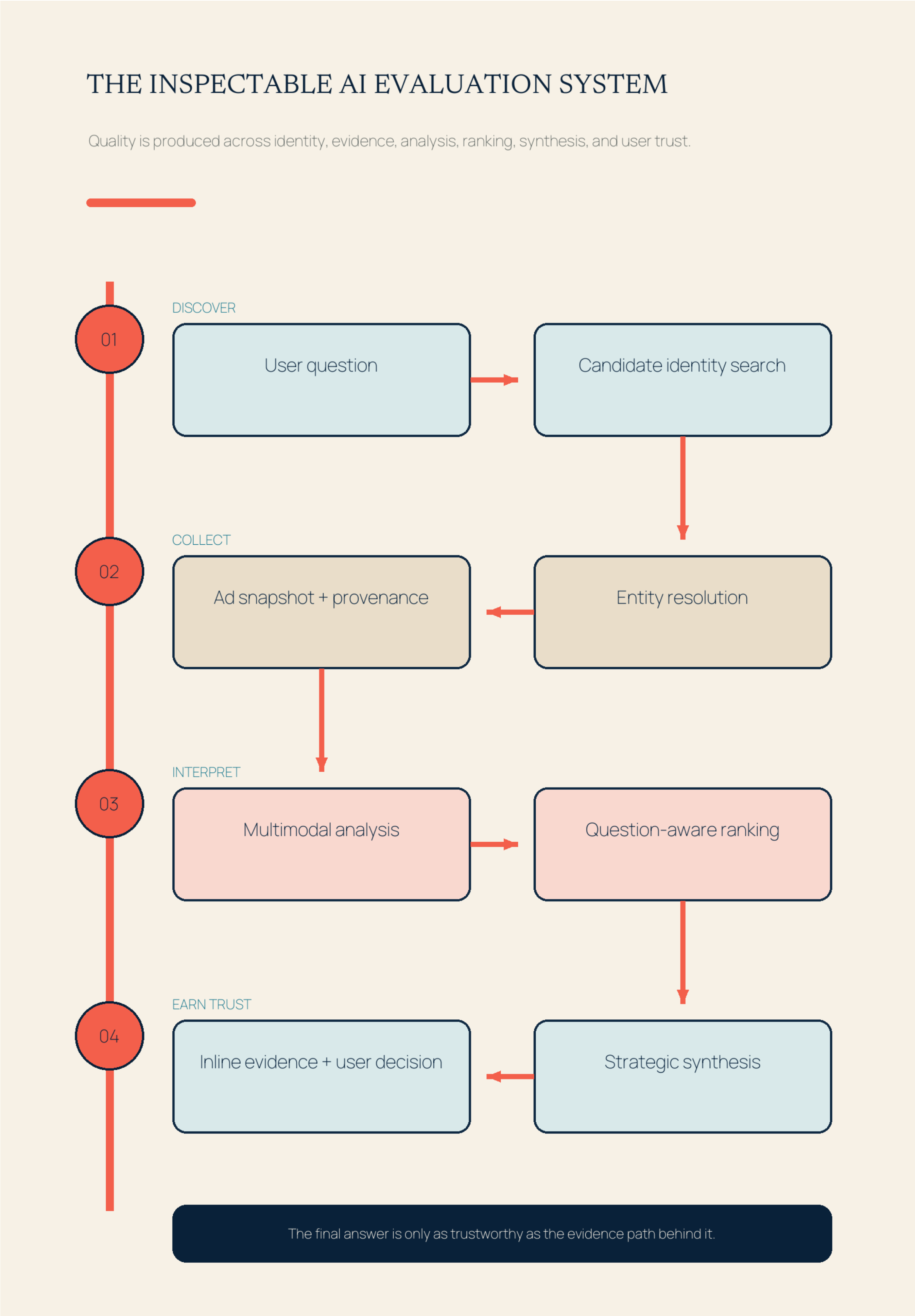
A hybrid design assigns exact control, bounded interpretation, validation, and user authority to different components.

Figure 12.2. Model-backed application pipeline



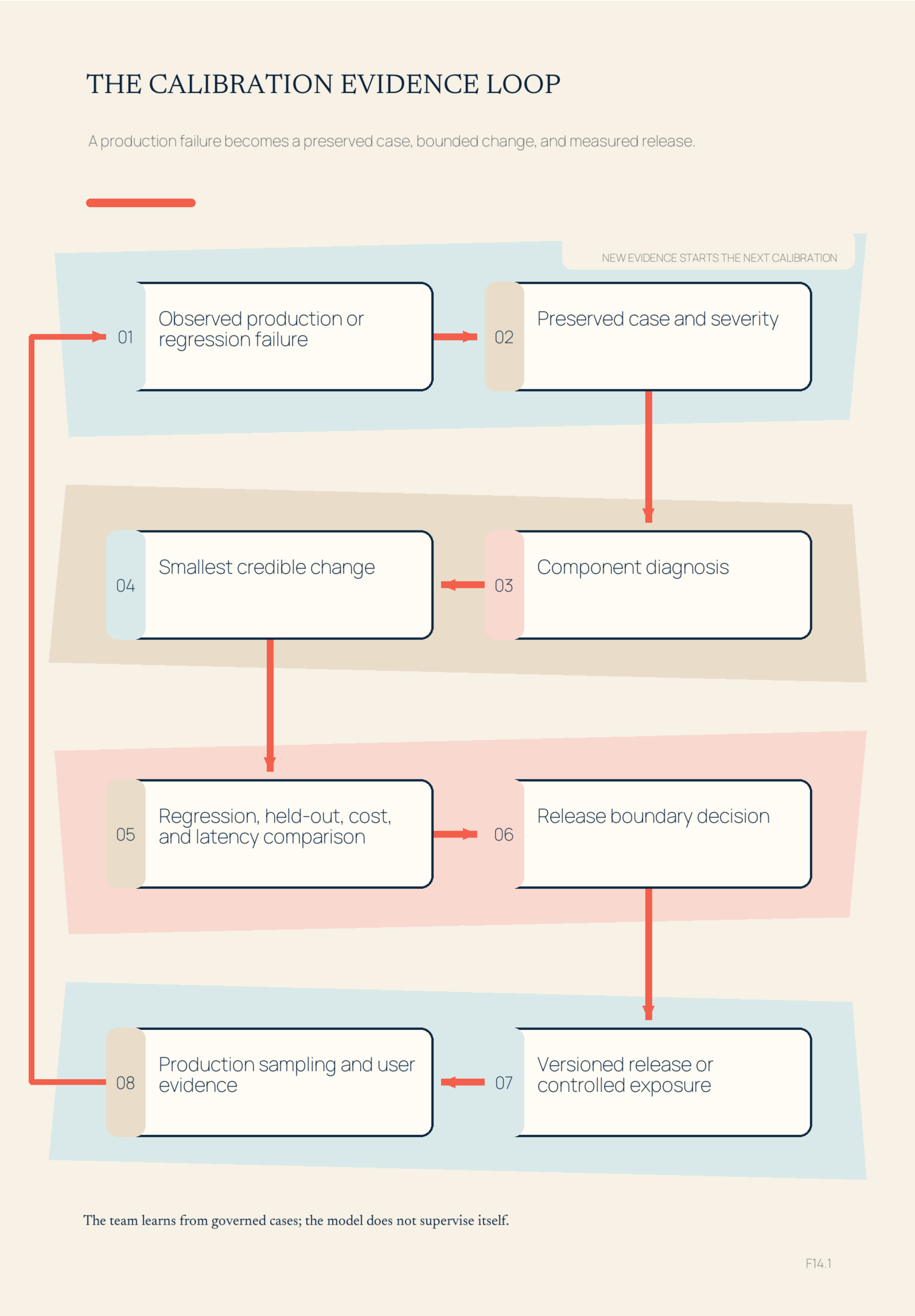
A model-backed feature is an application pipeline. Retrieval, context construction, generation, tool execution, validation, user control, tracing, and evaluation have different responsibilities.

Figure 13.1. Competitor-ad evaluation pipeline



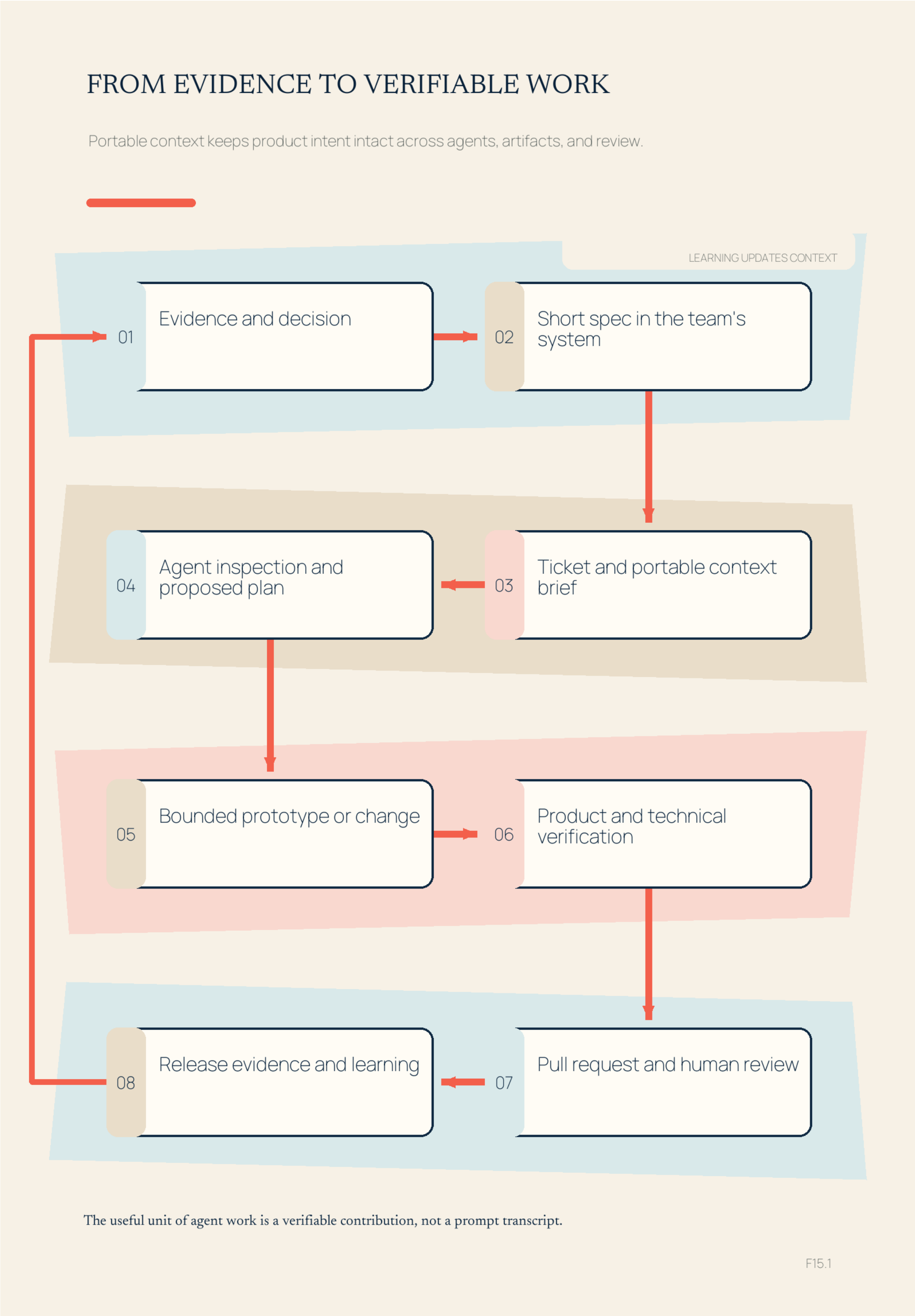
Final-answer quality depends on identity, evidence, analysis, ranking, synthesis, and UI trust—not only on the generation model.

Figure 14.1. Human-directed Heatseeker calibration loop



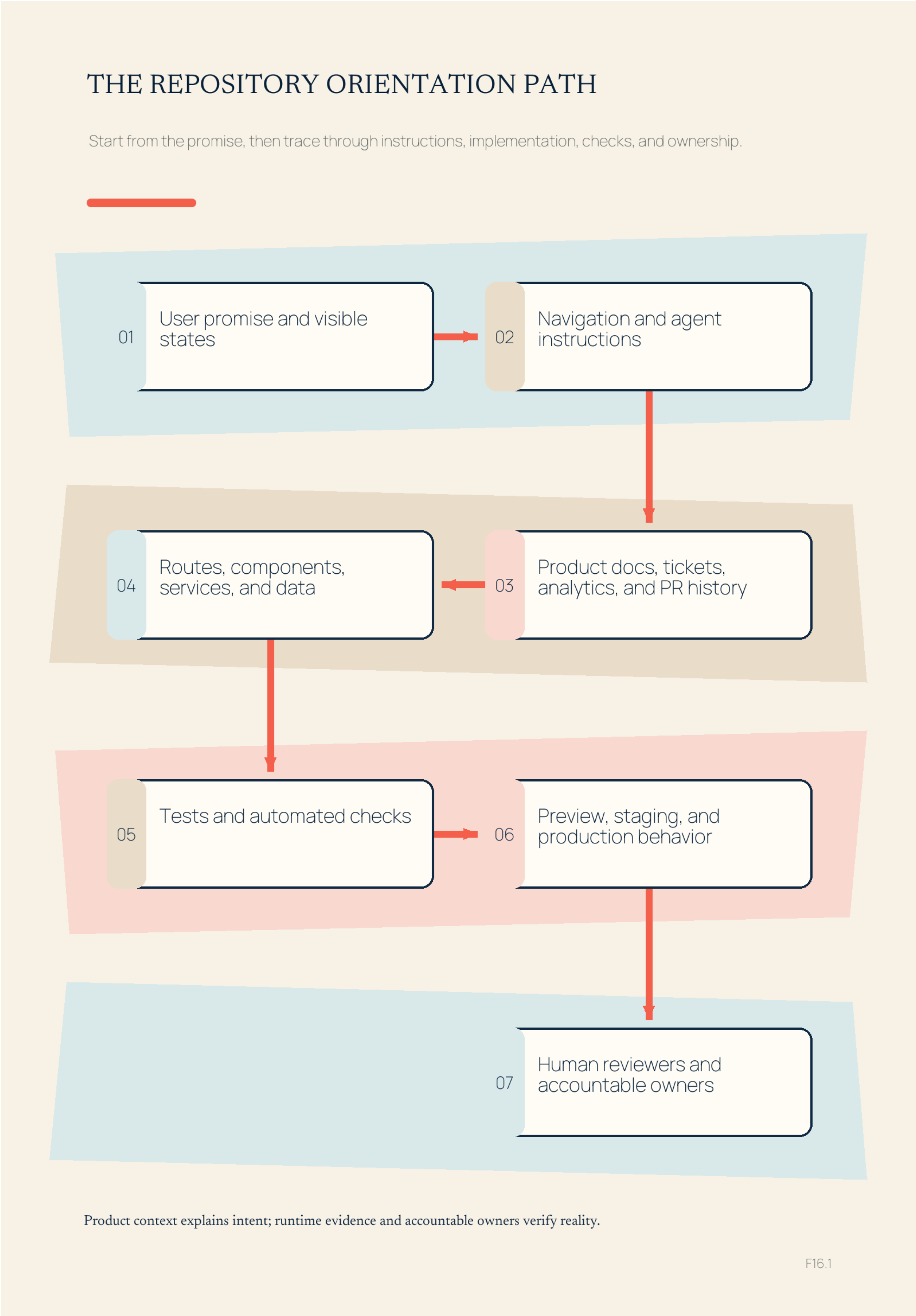
Calibration is a human-directed quality loop: an observed failure becomes a preserved case, bounded change, regression-tested release, and new production evidence.

Figure 15.1. Agent operating contract



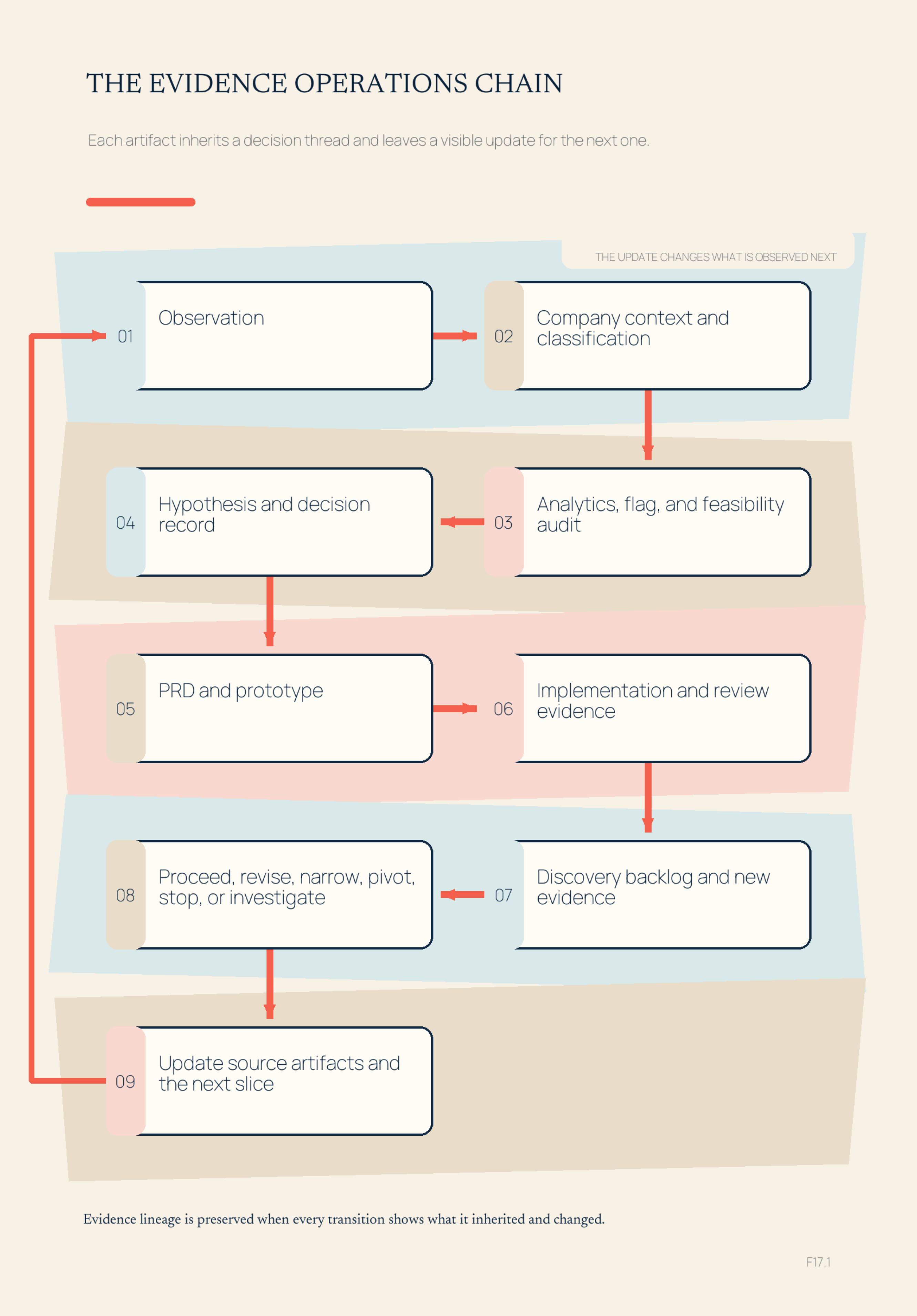
Agent context should preserve the evidence-to-decision thread and end in human-verifiable work, not an isolated prompt transcript.

Figure 16.1. Repo as product surface



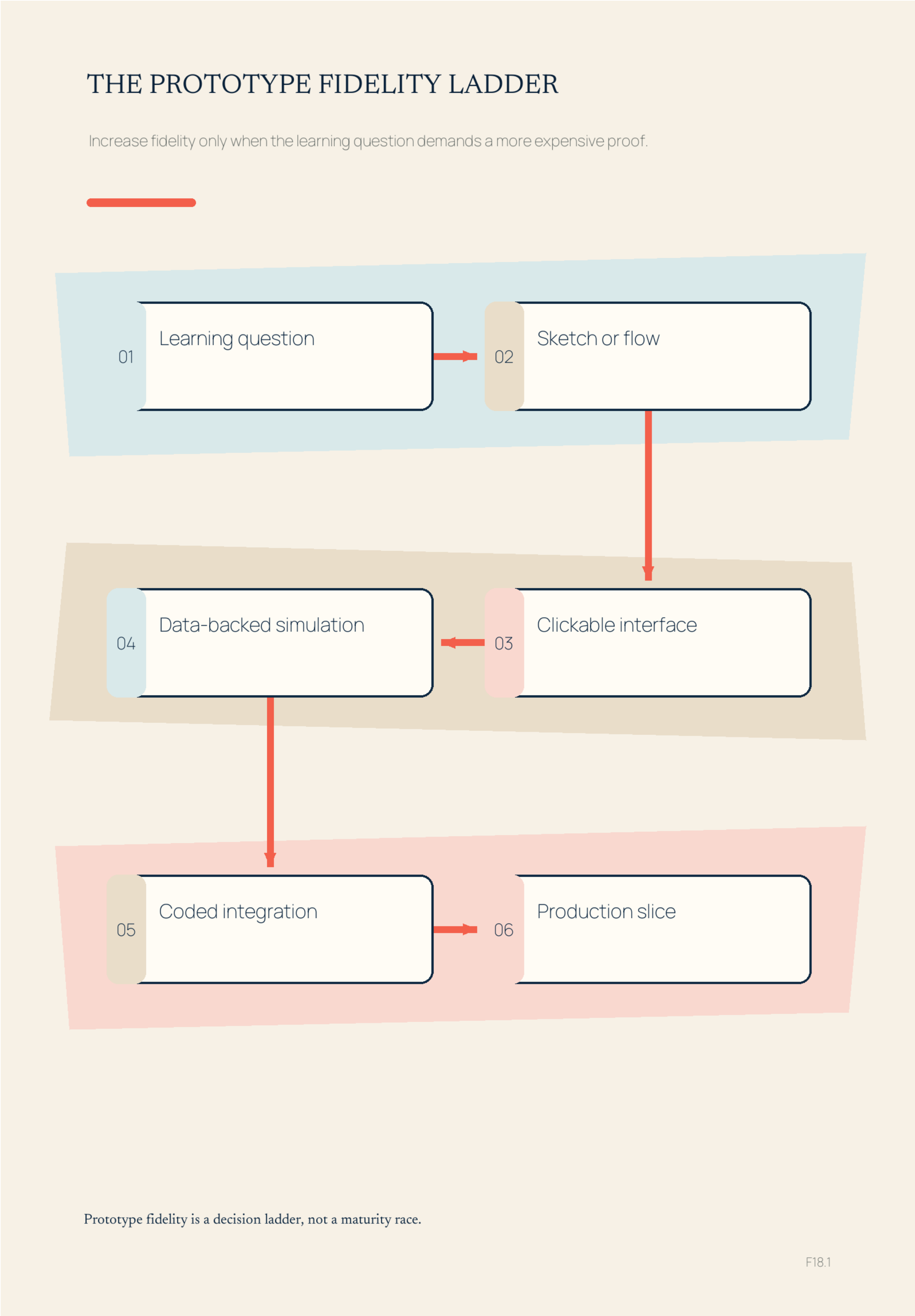
Repo orientation connects the user promise to implementation, checks, environments, and accountable reviewers; product context can explain rationale, but it does not replace runtime verification or human ownership.

Figure 17.1. Evidence operations continuity



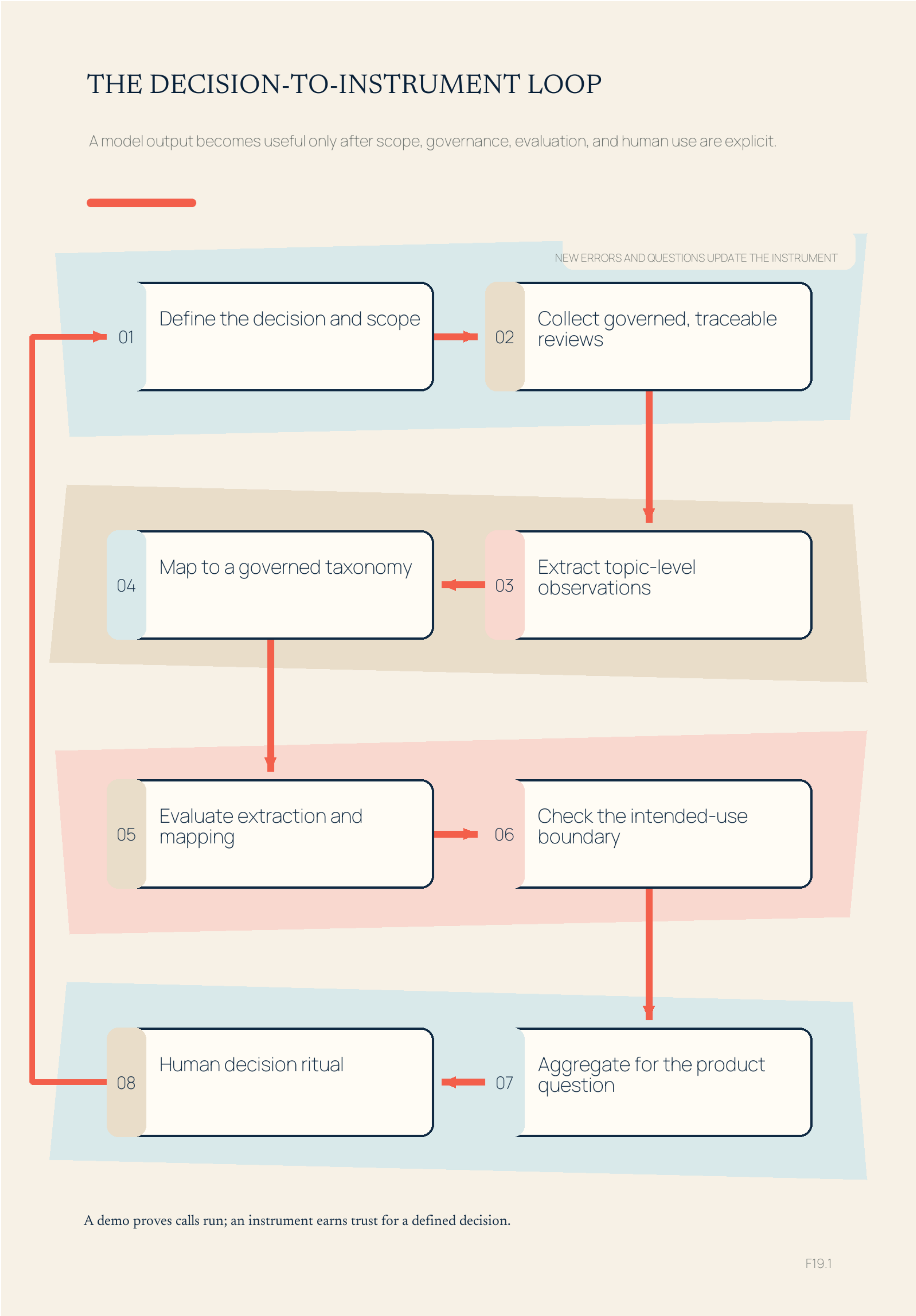
Evidence operations preserve lineage by carrying each observation through classification, product artifacts, implementation evidence, and an explicit decision that updates the next source of truth.

Figure 18.1. Prototype fidelity ladder



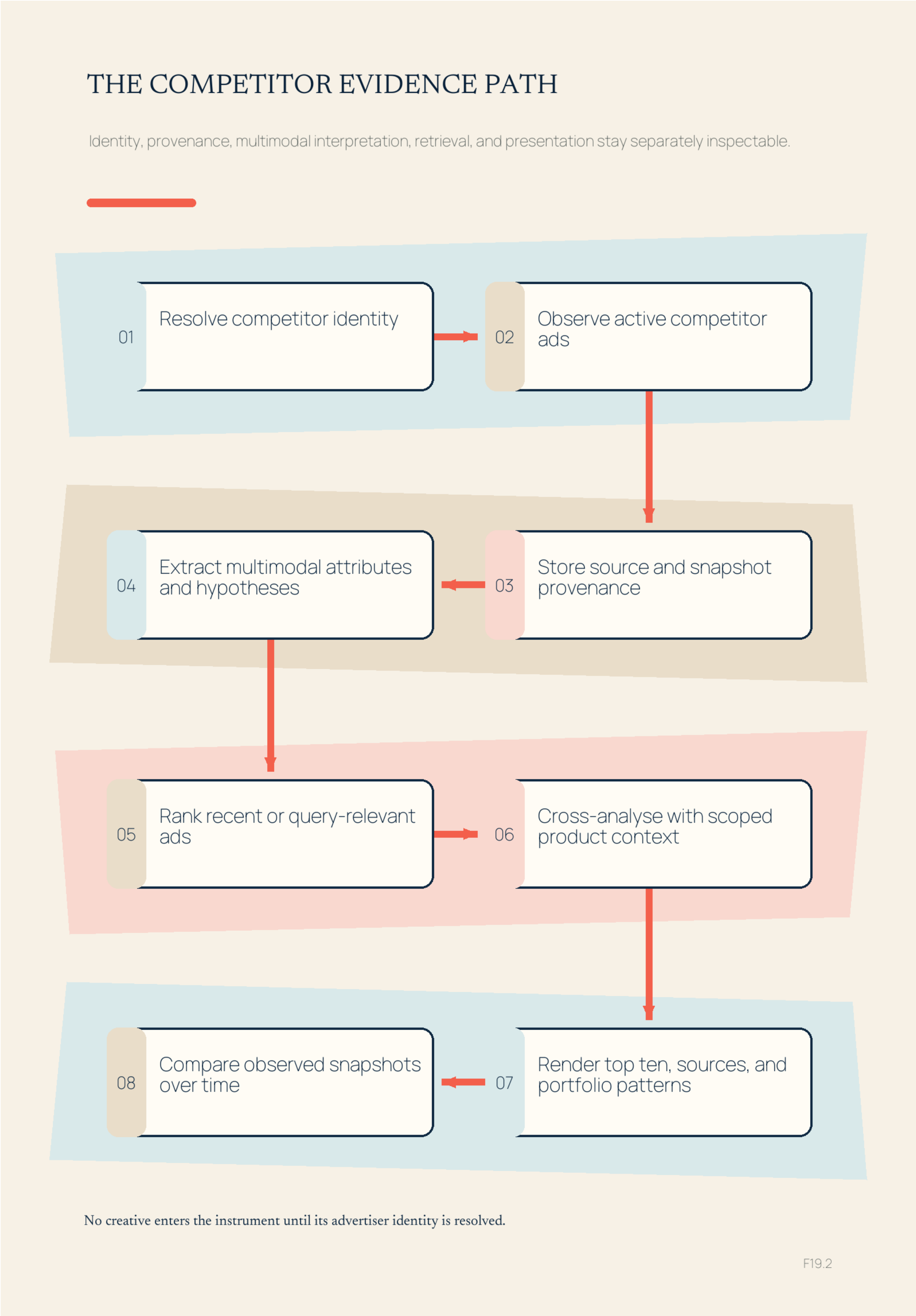
Prototype fidelity is a decision ladder, not a maturity race: move only as far as the learning question requires, and treat production exposure as a separate commitment.

Figure 19.1. Decision-to-instrument loop



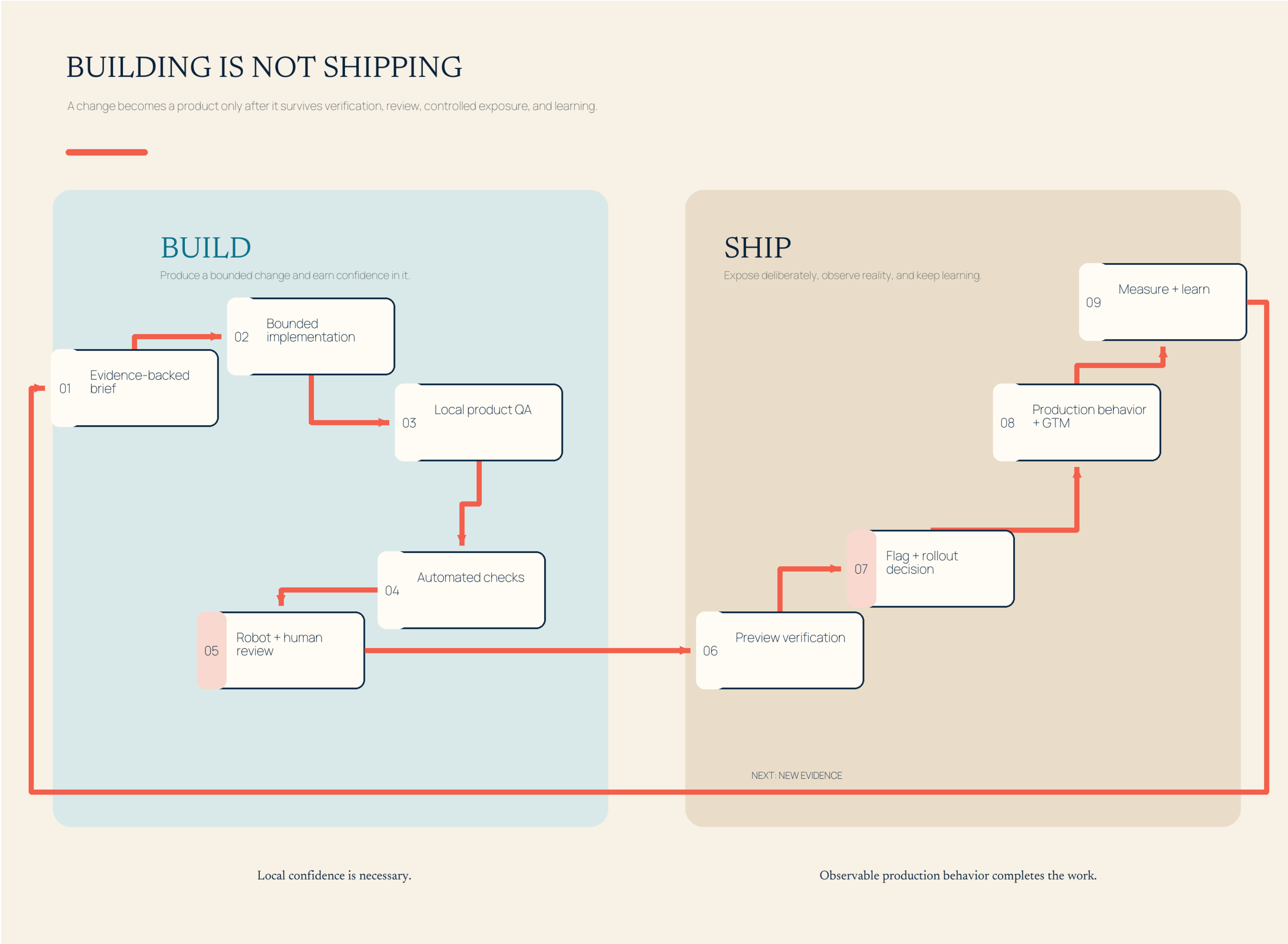
The original decision-to-instrument loop separates scope, governed collection, extraction, taxonomy, evaluation, aggregation, and human use so a model output does not become a dashboard without a quality boundary.

Figure 19.2. Competitor-ad transfer pipeline



A competitor-ad response is trustworthy only when identity resolution, collection, provenance, extraction, retrieval, cross-analysis, and presentation remain separately inspectable.

Figure 20.1. Building is not shipping



Building becomes shipping only after the change survives verification, review, controlled exposure, and measurement.